

PENCH ERP : LOGISTICS & DELIVERY

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TECHNICAL MANUAL

Comprehensive Guide for Implementation,
Operation, and Management



PENCH ERP

LOGISTICS & OPERATIONS TECHNICAL MANUAL

TECHNICAL DOCUMENTATION

Technical Documentation

Overview

Pench Backend is a multi-tenant ERP and Logistics platform built using Django. It supports multiple city-based schemas within a single database instance, providing isolation for city-level data while sharing core infrastructure.

Architecture

Multi-Tenancy

- > Framework: django-tenants
- > Isolation Strategy: Schema-based isolation.
- > Tenant Model: tenants.City
- > Domain Model: tenants.Domain
- > Shared Apps: accounts, tenants, core (Public Schema).
- > Tenant Apps: crm, orders, inventory, finance, hr, routing, tracking, administration (Tenant Schema).

Identity & Access Control (RBAC)

- > Centralized Identity: All users are stored in the public schema (accounts.User).
- > Roles:
 - > SuperAdmin: Unrestricted global access (Public & Tenants).
 - > Managers: Granular access to specific modules (Inventory, Logistics, HR, etc.) via Django Groups.
 - > Drivers: Mobile-app access for trip management and tracking.
 - > Customers: Limited access to their own orders and subscription portal.
- > Permission Classes: HasGroupPermission ensures API security based on group membership.

Authentication & Security

- > JWT: rest_framework_simplejwt for API authentication.

- > Token Lifetime: Access tokens are strictly valid for 1 day (24 hours).
- > OTP: Custom OTP model for secure phone-based logins (Drivers/Customers).
- > API Metadata: Every response includes `expires_in_seconds` for session management.

Complete Operational Flow

The following describes the end-to-end lifecycle of the Pench delivery ecosystem:

1. Tenant & System Initialization

- > City Setup: SuperAdmin creates a new City (Tenant) and links a Domain.
- > Role Sync: The system automatically initializes standard Groups (SuperAdmin, Staff, Manager, Driver, Customer) on first run.
- > Zone Definition: Managers define geographic zones (PolygonField) for delivery coverage.

2. User Onboarding & CRM

- > Lead Generation: Potential customers are added as Leads in the crm module.
- > Conversion: Leads are converted to Customers with precise GIS location pinning.
- > Identity Link: Each customer is linked to a global User account for portal access.

3. Subscription & Order Generation

- > Plan Assignment: Customers subscribe to specific product plans (e.g., Daily Milk).
- > Auto-Generation: Orders are automatically generated based on the subscription frequency or manually created by Managers.
- > Inventory Check: System verifies product availability across warehouses.

4. Logistics & Route Optimization

- > Trip Creation: Orders are grouped into Trips for a specific delivery date.
- > Optimization: The routing engine uses Google OR-Tools and OSRM to generate the most efficient delivery sequence (TSP).
- > Driver Assignment: Trips are assigned to specific Drivers based on availability.

5. Driver Execution (Mobile App)

- > Login: Drivers authenticate via OTP.
- > Route Tracking: The app uses a WebView-based Leaflet integration to display the optimized route and live location.
- > Live Streaming: Real-time GPS coordinates are streamed to the backend for monitoring.

6. Delivery Fulfillment

- > QR Verification: Upon arrival, the driver must scan a QR code at the delivery point to verify location.
- > POD Upload: Proof of Delivery (POD) photos are uploaded as evidence.
- > Atomic Transaction: Once "Delivered" is triggered:
 1. Order status updates to delivered.
 2. Inventory is deducted (including bottle returns).
 3. Invoice is automatically generated in the finance module.

App Descriptions

accounts

Handles user registration, authentication, and global RBAC permissions.

tenants

Defines cities (tenants) and domains. Also manages geographic zones.

orders

Manages order lifecycle, delivery schedules, and trip groupings.

inventory

Tracks products, categories, and returnable assets (bottles).

routing & tracking

Handles logistics optimization, GPS events, and live tracking visualization.

administration

Tenant-specific configurations (Theme, POD requirements, support).

Deployment

- > Containerization: Docker / Docker Compose.
- > Web Server: Daphne (ASGI) for WebSockets and WSGI for REST.

PROJECT MILESTONES

Project Milestones

This document tracks the major milestones completed in the PENCH Backend project.

Completed Milestones

1. Core Architecture & Multi-Tenancy

- > [x] Initialized Django project with django-tenants.
- > [x] Configured shared vs. tenant app separation.
- > [x] Implemented City/Domain routing for dynamic schema switching.
- > [x] Setup PostgreSQL with PostGIS for spatial data support.
- > [x] [ENHANCED] Frontend multi-tenancy: Dynamic URL routing and CORS-aware proxying.

2. Authentication & Identity Management

- > [x] [ENHANCED] Global Identity Model: All users (Admins, Drivers, Customers) managed centrally in the Public schema.
- > [x] Phone-based OTP authentication system with rest_framework_simplejwt.
- > [x] JWT integration with 1-day token validity and expires_in_seconds metadata.
- > [x] [NEW] Automated Role Setup: Server-startup group initialization (SuperAdmin, Managers, Drivers, Customers).
- > [x] [NEW] Enhanced Serializers: Mandatory phone and email validation for robust identity tracking.

3. CRM & Customer Management

- > [x] Lead management and customer profiling with GIS locations.
- > [x] Subscription plans and automated role assignment.
- > [x] [NEW] Customer Portal: Responsive interface for order management and subscription tracking.

4. Logistics & Optimization

- > [x] Route optimization using Google OR-Tools (TSP/VRP) and OSRM driving distance matrix.

- > [x] Real-time tracking events with GeoJSON geometry generation.
- > [x] [ENHANCED] Live Tracking: Migrated to secure WebView-based Leaflet integration for cross-platform reliability.

5. Delivery Fulfillment & Evidence

- > [x] [NEW] Driver Mobile Flow: Integrated route management and delivery status updates.
- > [x] [NEW] QR Code Scanning: Mandatory scan at delivery point for location verification.
- > [x] [NEW] Proof of Delivery (POD): Multi-photo upload requirement with atomic state transitions.
- > [x] [NEW] Atomic Fulfillment: Synchronized order completion, stock deduction, and auto-invoicing.

6. Inventory & Assets

- > [x] Product and Category management.
- > [x] Reusable bottle tracking (Issued vs. Returned).
- > [x] Automated stock adjustment upon delivery fulfillment.

7. Documentation & Tooling

- > [x] [NEW] Postman Collection v4.2.1: Role-based API documentation with standardized environments.
- > [x] [NEW] Technical Driver Manual: Comprehensive guide for mobile application workflows.
- > [x] [NEW] Migration Safety: Detailed protocols for schema migrations and data integrity.

Current Focus

- > [] Advanced Finance & Billing reconciliation dashboards.
- > [] Push Notifications for Order Status and Delivery Alerts.
- > [] AI-driven demand forecasting based on subscription patterns.

ADMIN MANUAL



Project Manual: Admin

Welcome to the Pench ERP Administration Manual. This guide explains how to manage your city/tenant and configure system behavior.

1. Dashboard Overview

The Admin dashboard provides a summary of:

- > Active orders for the day.

- > Driver statuses (On-duty vs. Off-duty).
- > Critical inventory levels.

2. Configuration Settings

Navigate to Administration -> Admin Configuration to manage:

- > Enable Delivery Photo: If on, drivers MUST upload a photo to complete delivery.
- > Auto-Assign Orders: Automatically assigns orders to drivers based on historical routes.
- > Max Cancellation Time: Minutes before delivery start when customers can no longer cancel.
- > Theme Customization: Update your company name and primary theme color.

3. User & Identity Management

All users are managed under User Management in the Public Schema:

- > Global Identity: Admins, Drivers, and Customers are centrally stored.
- > Automated Roles: Groups (SuperAdmin, Manager, Driver, Customer) are initialized automatically.
- > Permissions: Use Django Groups for RBAC (Role Based Access Control).

4. Developer Tools

- > API Documentation: A role-based Postman collection (v4.2.1) is available in the documentation/ folder for system integration.
- > Token Monitoring: All API responses include expires_in_seconds for session tracking.

5. Route Optimization

To create a route:

1. Go to Logistics -> Optimized Routes.
2. Select the delivery date and the orders to include.
3. Click "Generate Optimized Route".
4. Assign a driver to the route.

6. Inventory Control

- > Products: Add new products and set categories.
- > Bottle Tracking: Monitor "Issued" vs. "Returned" bottles for each customer to ensure asset recovery.

DRIVER MANUAL



Project Manual: Driver

Welcome to the Pench Delivery Mobile Manual. This guide explains how to use the app to fulfill your daily deliveries.

1. Getting Started

- > Login: Use your registered phone number. You will receive an OTP via SMS.
- > My Route: Once logged in, tap "My Route" to see your assigned deliveries for today.

2. Starting a Trip

1. Open your assigned route.
2. Tap "Start Trip". This notifies the customers that you are on your way.
3. The map will show the optimized sequence of stops.

3. Completing a Delivery

For each stop:

1. Arrive at the customer location.
2. QR Verification: Tap "Scan QR" and point your camera at the customer's unique QR code to verify your arrival.
3. Tap "Submit Delivery".
4. Proof of Delivery (POD): If enabled, you must take a photo of the delivered items at the customer's door.
5. Bottle Return: Enter the number of empty bottles you collected from the customer.
6. Tap "Confirm".

4. Finishing the Day

- > Once all stops are completed, tap "Complete Trip".
- > Your performance stats (distance covered, bottles collected) will be updated automatically.

5. Support

If you face issues with a location or an order:

- > Tap the "Call Support" button in the app.
- > Or use the internal chat to message the ERP admin.

CUSTOMER MANUAL

Project Manual: Customer

Welcome to the Pench Customer Portal Manual. This guide explains how to manage your subscriptions and orders.

1. Login

- > Use your registered mobile number and verify via OTP.

2. Managing Subscriptions

- > View your active subscription plans (e.g., Daily Milk, Alternate Day Water).
- > Pause/Resume: Pause your delivery if you are away.
- > Modify: Change the quantity of items for your next delivery.

3. Placing One-Time Orders

- > Browse the product catalog.
- > Add items to your cart.
- > Select a delivery date and pay securely.

4. Tracking & Notifications

- > Receive an SMS/Push notification when the driver starts the trip.
- > View real-time tracking of the delivery vehicle on the map.
- > Get a confirmation photo once the item is delivered.

5. Billing & Payments

- > View your ledger and transaction history.
- > Download invoices for past orders.
- > Pay outstanding balances via integrated payment gateways (Razorpay/Paytm).